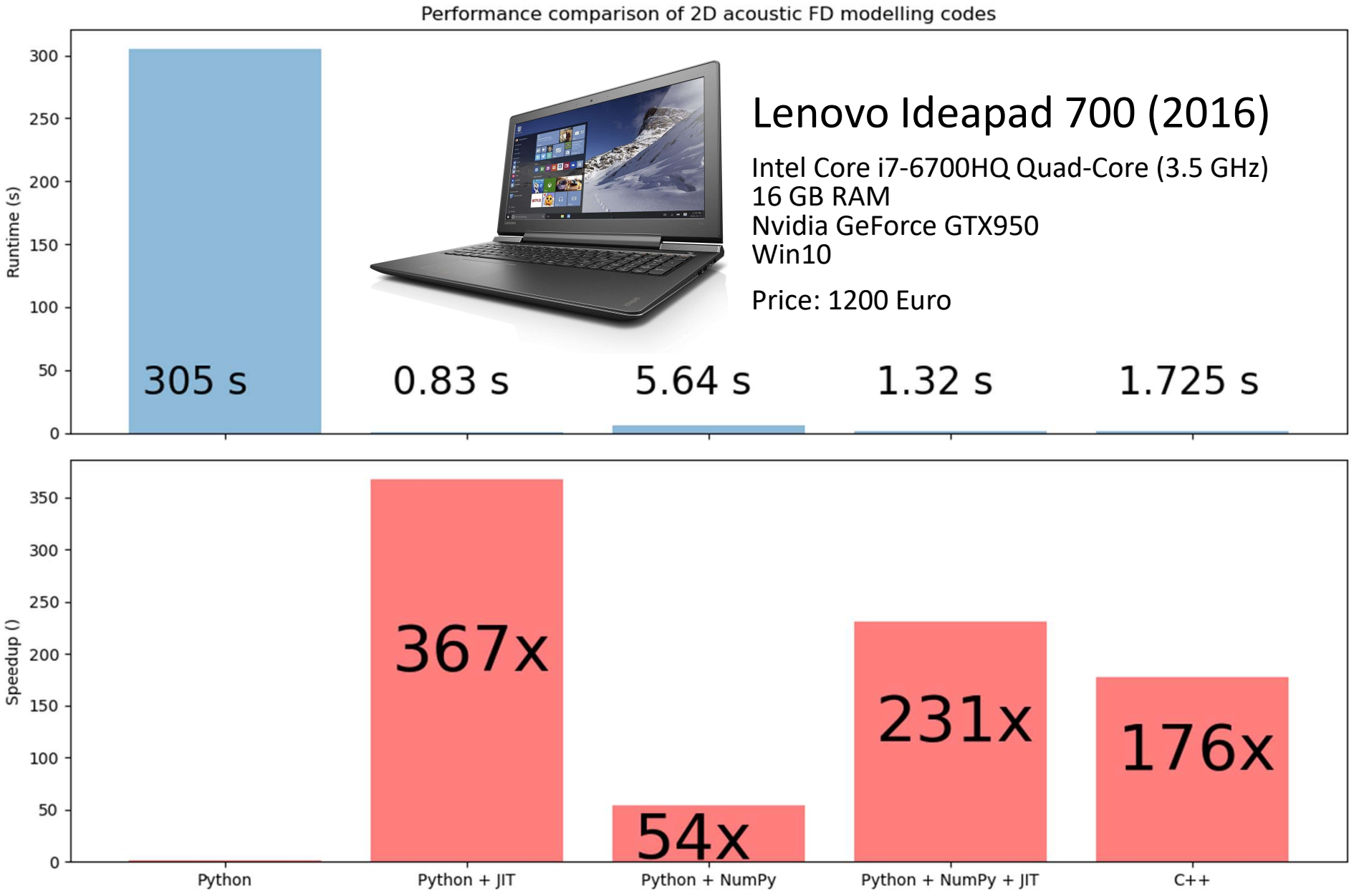
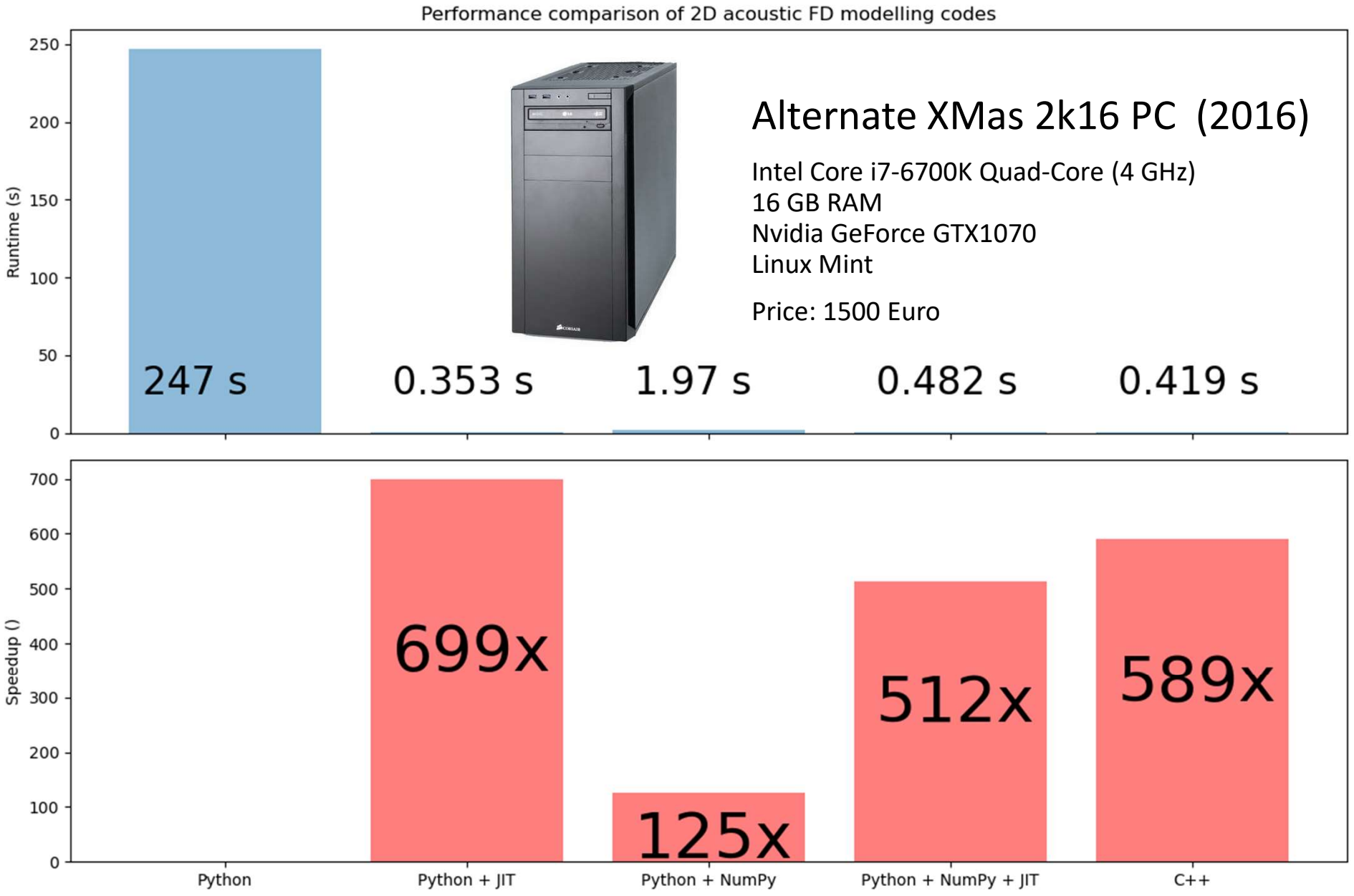
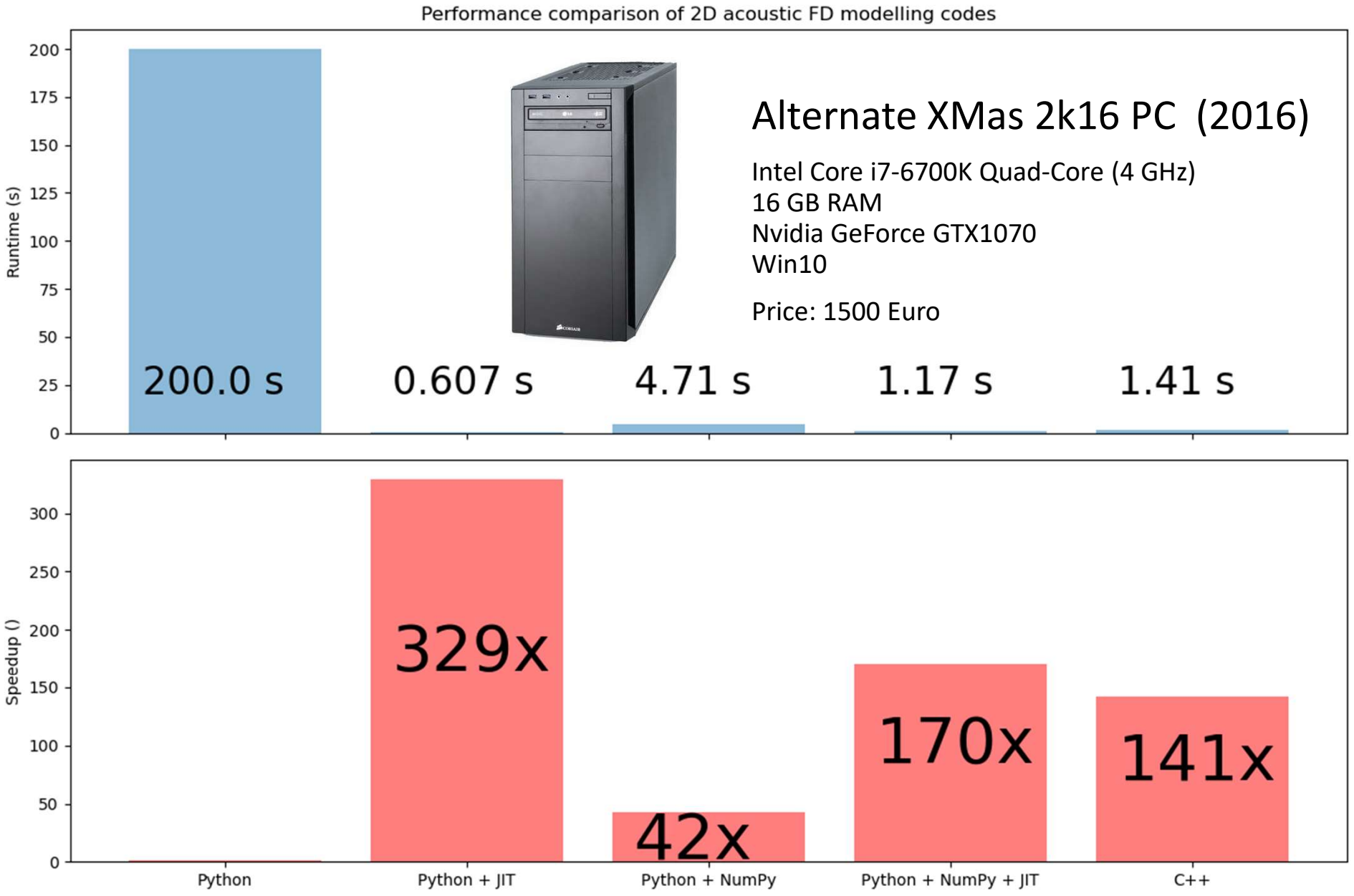
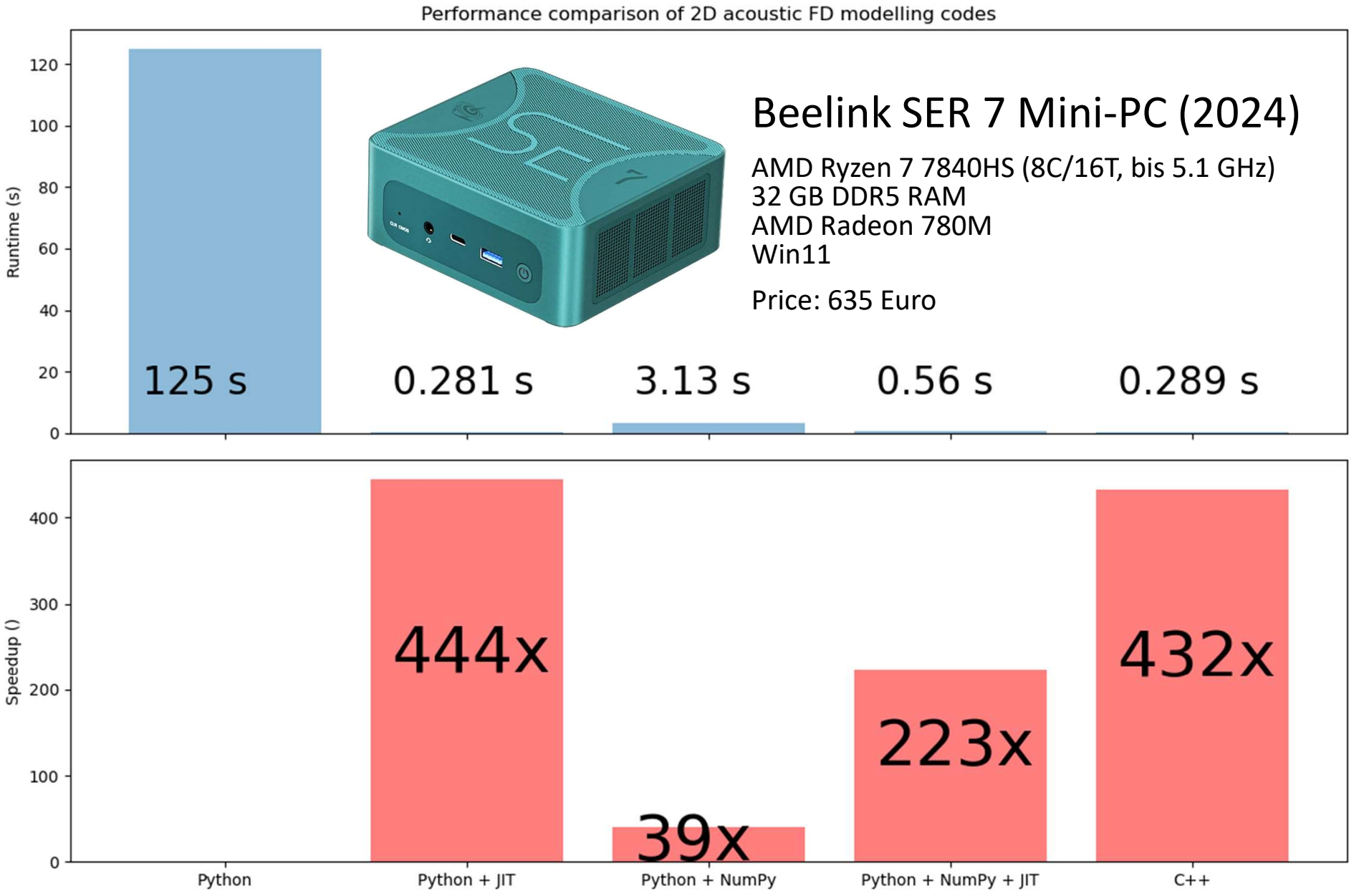
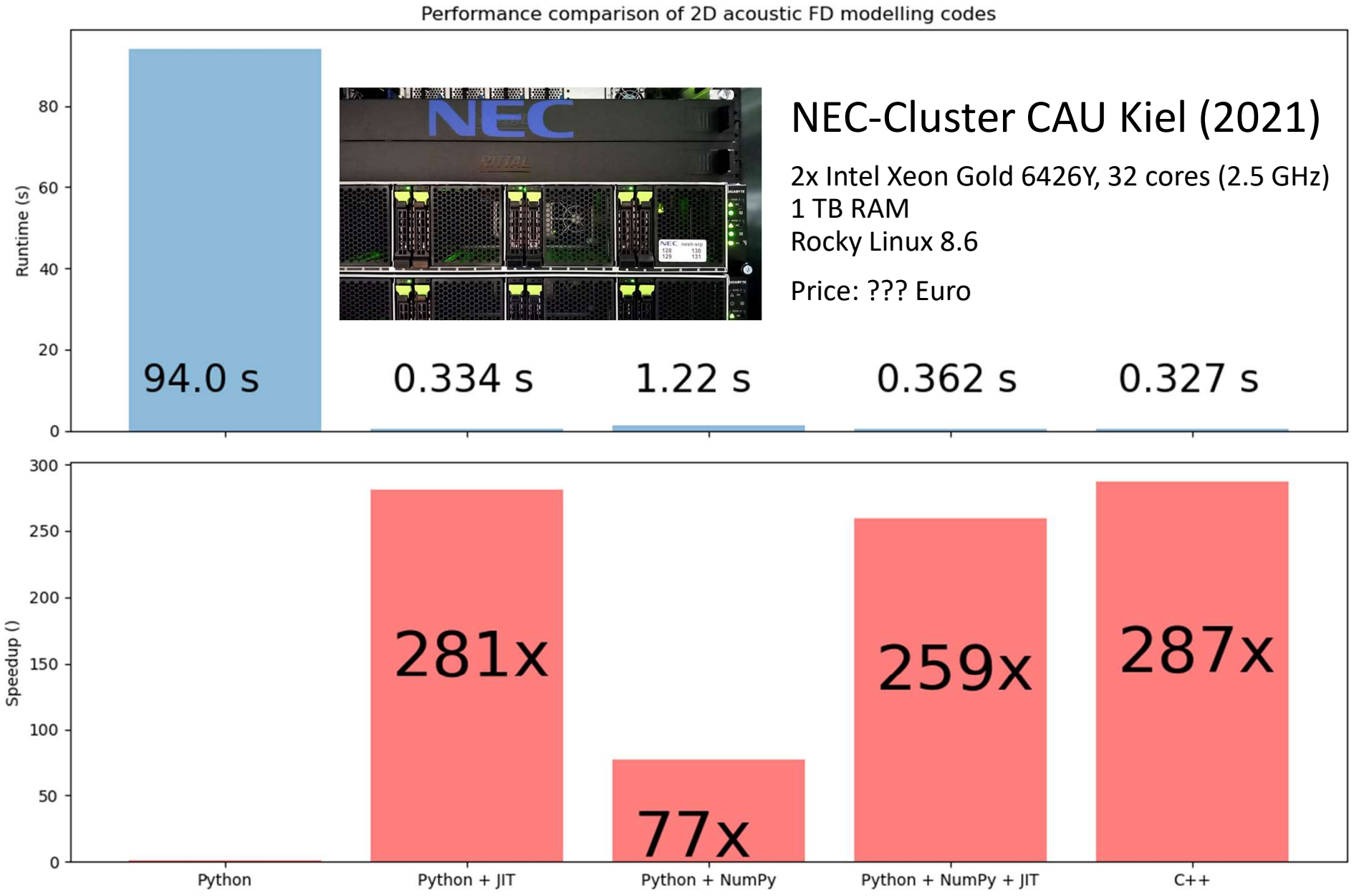
Benchmark Results
2D acoustic Modelling Code


Performance comparison of 2D acoustic FD modelling codes
Runtime (s)
305 s
0.83 s
5.64 s
1.32 s
1.725 s
Lenovo Ideapad 700 (2016)
Intel Core i7-6700HQ Quad-Core (3.5 GHz)
16 GB RAM
Nvidia GeForce GTX950
Win10
Price: 1200 Euro
Speedup ()
367x
54x
231x
176x
Python
Python + JIT
Python + NumPy
Python + NumPy + JIT
C++


Performance comparison of 2D acoustic FD modelling codes
Runtime (s)
250
200
150
100
50
0
247 s
0.353 s
1.97 s
0.482 s
0.419 s
Alternate XMas 2k16 PC (2016)
Intel Core i7-6700K Quad-Core (4 GHz)
16 GB RAM
Nvidia GeForce GTX1070
Linux Mint
Price: 1500 Euro
Speedup ()
700
600
500
400
300
200
100
0
699x
125x
512x
589x
Python
Python + JIT
Python + NumPy
Python + NumPy + JIT
C++


Performance comparison of 2D acoustic FD modelling codes
Runtime (s)
200
175
150
125
100
75
50
25
0
200.0 s
0.607 s
4.71 s
1.17 s
1.41 s
Alternate XMas 2k16 PC (2016)
Intel Core i7-6700K Quad-Core (4 GHz)
16 GB RAM
Nvidia GeForce GTX1070
Win10
Price: 1500 Euro
Speedup ()
300
250
200
150
100
50
0
329x
42x
170x
141x
Python
Python + JIT
Python + NumPy
Python + NumPy + JIT
C++


Performance comparison of 2D acoustic FD modelling codes
Runtime (s)
120
100
80
60
40
20
0
125 s
0.281 s
3.13 s
0.56 s
0.289 s
Beelink SER 7 Mini-PC (2024)
AMD Ryzen 7 7840HS (8C/16T, bis 5.1 GHz)
32 GB DDR5 RAM
AMD Radeon 780M
Win11
Price: 635 Euro
Speedup ()
400
300
200
100
0
444x
39x
223x
432x
Python
Python + JIT
Python + NumPy
Python + NumPy + JIT
C++


Performance comparison of 2D acoustic FD modelling codes
Runtime (s)
0
20
40
60
80
94.0 s
0.334 s
1.22 s
0.362 s
0.327 s
NEC-Cluster CAU Kiel (2021)
2x Intel Xeon Gold 6426Y, 32 cores (2.5 GHz)
1 TB RAM
Rocky Linux 8.6
Price: ??? Euro
Speedup ()
0
50
100
150
200
250
300
281x
77x
259x
287x
Python
Python + JIT
Python + NumPy
Python + NumPy + JIT
C++